

7.1 Philosophical Motivations

One of its most distinctive features of S_0 is its treatment of contradictions: once a contradiction is admitted, any formula follows. This consequence is expressed by the principle *ex contradictione quodlibet* (ECQ), also known as the principle of explosion:

$$A, \neg A \text{ / } B. \quad (\text{ECQ})$$

As a result, any inconsistent set of or sequence of formulae becomes trivial, in the sense that every formula follows from it by (ECQ).

The idea that contradictions leads to triviality has been used by many authors (e.g., Popper) to discard inconsistencies from science and knowledge in general. This idea has nevertheless been challenged on both philosophical and practical grounds.

A source of motivation comes from the study of semantic paradoxes. An example is the liar's paradox, generated by a sentence like:

$$(\neg) \text{ is false.} \quad (\neg)$$

This sentence leads to a contradiction and, hence, to a triviality if reasoned by means of classical logic. However, there does not seem to be anything ungrammatical about it. So it has the form of a propositional sentence – although it is arguably not.

Similar difficulties arise in set theory. Naïve set theory, based on unrestricted comprehension, gives rise to Russell's paradox and other antinomies. Such paradoxes suggest that contradictions may arise naturally within formal systems and that their presence need not imply the collapse of all reasoning.

Another important motivation is the existence of inconsistent information. Databases, legal systems, regulations, and large bodies of knowledge often contain conflicting statements. Such inconsistencies may result from errors, temporal changes, or the integration of data from multiple sources. Under classical logic, it seems impossible to reason about those databases, as they would be trivial. In practice, however, people are able to draw sensible conclusions while avoiding arbitrary ones.

The history of science provides further examples. Scientific theories have often contained tensions and even outright contradictions. Yet scientists have continued to employ such theories successfully, extracting valuable predictions and explanations despite these inconsistencies. The development of science therefore suggests that inconsistency and rationality are not incompatible. This also

suggests a non-classical approach where not every contradiction leads to triviality.

These considerations motivate the development of *paraconsistent logics*. A logic is paraconsistent (with respect to the negation operator ‘ $\neg$ ’) when the (ECQ) does not hold in general. Such logics make possible the formal study of inconsistent but non-trivial systems (e.g., theories, databases, etc.).

Among the most influential systems of paraconsistent logic are those introduced by Newton da Costa (1963). His calculi, which we will call $\mathsf{C0}_n$ (where $n \in \mathbf{Num}_{\mathbb{N}}$)*, constituted the first systematic family of paraconsistent propositional logics explicitly intended to solve semantic paradoxes by accepting contradictions.

*: Originally called C_n , where $n \in \mathbf{Num}_{\mathbb{N}} \cup \{\omega\}$.

7.2 Vocabularies and Languages

The vocabularies and formal languages $\mathbf{Vo}_{\mathsf{C0}_1}, \dots$ and $\mathbf{Fo}_{\mathsf{C0}_1}, \dots$ will be defined as those of $\mathsf{S0}$.

Definition 7.1.

$$\mathbf{Fo}_{\mathsf{C0}_n} \stackrel{\text{DF}}{=} \mathbf{Fo}_{\mathsf{S0}}.$$

$$\mathbf{Vo}_{\mathsf{C0}_n} \stackrel{\text{DF}}{=} \mathbf{Vo}_{\mathsf{S0}}.$$

In order to better express some principles of these systems, we will use the following abbreviations:

Definition 7.2 (Abbreviations for $\mathbf{Fo}_{\mathsf{C0}_1}, \dots$).

$$(A)^1 \stackrel{\text{DF}}{=} \neg(A \wedge \neg A)$$

$$(A)^{N(m)} \stackrel{\text{DF}}{=} ((A)^m)^1$$

$$A^{(1)} \stackrel{\text{DF}}{=} A^1$$

$$(A)^{(N(m))} \stackrel{\text{DF}}{=} ((A)^{(m)} \wedge (A)^{N(m)})$$

$$\neg^{(m)} A \stackrel{\text{DF}}{=} (\neg A \wedge (A)^{(m)})$$

The expressions defined above are not formulae of $\mathbf{Fo}_{\mathsf{C0}_1}, \dots$, but rather abbreviations of some formulae of them. We will use these abbreviations in our meta-language in order to more easily describe the properties of the systems $\mathsf{C0}_1, \dots$ by da Costa. We will dispense with the parentheses when this does not result in any ambiguity.

Thus, the expression ‘ p^1 ’, which abbreviates the formula ‘ $\neg(p \wedge \neg p)$ ’, is understood by da Costa to intuitively express that ‘ p ’ is *well-behaved*. One might also think of it as stating that ‘ p ’ is consistent, that is, that it is not the case that both ‘ p ’ and ‘ $\neg p$ ’. A formula is well-behaved when it is not simultaneously true or asserted together with its negation.

The expression ' p^2 ', which abbreviates the expression ' $\neg(p^1 \wedge \neg p^1)$ ', which in turn abbreviates the formula ' $\neg(\neg(p \wedge \neg p) \wedge \neg\neg(p \wedge \neg p))$ ', has the intended sense that ' $\neg(p \wedge \neg p)$ ' is well-behaved (though not necessarily also ' p '). That is, not only ' p ' is well-behaved, but also the formulae expressing that it is well-behaved and that is ill-behaved – that is, the formula $(p \wedge \neg p)$.

An expression of the form ' $\ulcorner p^{(m)} \urcorner$ ' abbreviates a formula of the form ' $\ulcorner p^1 \wedge \dots \wedge p^m \urcorner$ ' (if we remove parentheses). The intuitive sense is that ' p ' is well-behaved up to level m . In other words, not only is ' p ' well-behaved, but so are ' p^1 ' and ' p^2 ' and so forth until ' p^m '.

Finally, the expression ' $\overset{(1)}{\neg} p$ ', which abbreviates ' $(\neg p \wedge p^{(1)})$ ', has the intended sense of strong negation in the system $\mathbf{C0}_1$. It is strong because, as we will see, although ' $(\neg p \wedge p^{(1)})$ ' does not lead to explosion in $\mathbf{C0}_1$, the formula abbreviated by ' $(p \wedge \overset{(1)}{\neg} p)$ ' does. In general, for all $n \in \mathbf{Num}$, the symbol ' $\ulcorner \overset{(n)}{\neg} \urcorner$ ' expresses the operation of strong negation in the system called ' $\mathbf{C0}_n$ '.

Task 7.1. Determine the formulae abbreviated by the following expressions:

1. p^2
2. $(p \wedge q)^1$
3. $(p \wedge q)^2$
4. $p^{(1)}$
5. $p^{(2)}$
6. $\overset{(1)}{\neg} p$
7. $\overset{(2)}{\neg} p$
8. $\overset{(1)}{\neg}(p \wedge \overset{(1)}{\neg} p)$

Solution to task 7.1.

1. $p^2 = (p^1)^1$
 $= \neg(p^1 \wedge \neg p^1)$
 $= \neg(\neg(p \wedge \neg p) \wedge \neg\neg(p \wedge \neg p))$
2. $(p \wedge q)^1 = \neg((p \wedge q) \wedge \neg(p \wedge q))$
3. $(p \wedge q)^2 = ((p \wedge q)^1)^1$
 $= \neg((p \wedge q)^1 \wedge \neg(p \wedge q)^1)$
 $= \neg(\neg((p \wedge q) \wedge \neg(p \wedge q)) \wedge \neg\neg((p \wedge q) \wedge \neg(p \wedge q)))$
4. $p^{(1)} = p^1$
 $= \neg(p \wedge \neg p)$
5. $p^{(2)} = (p^{(1)} \wedge p^2)$
 $= (p^1 \wedge p^2)$
 $= (\neg(p \wedge \neg p) \wedge \neg(\neg(p \wedge \neg p) \wedge \neg\neg(p \wedge \neg p)))$
6. $\overset{(1)}{\neg} p = (\neg p \wedge p^{(1)})$
 $= (\neg p \wedge p^1)$
 $= (\neg p \wedge \neg(p \wedge \neg p))$
7. $\overset{(2)}{\neg} p = (\neg p \wedge p^{(2)})$
 $= (\neg p \wedge (p^1 \wedge p^2))$
 $= (\neg p \wedge (\neg(p \wedge \neg p) \wedge \neg(\neg(p \wedge \neg p) \wedge \neg\neg(p \wedge \neg p))))$
8. $\overset{(1)}{\neg}(p \wedge \overset{(1)}{\neg} p) = (\neg(p \wedge \overset{(1)}{\neg} p) \wedge (p \wedge \overset{(1)}{\neg} p))$
 $= (\neg(p \wedge \overset{(1)}{\neg} p) \wedge (p \wedge \overset{(1)}{\neg} p))$
 $= (\neg(p \wedge (\neg p \wedge p^{(1)})) \wedge (p \wedge (\neg p \wedge p^{(1)})))$
 $= (\neg(p \wedge (\neg p \wedge p^1)) \wedge (p \wedge (\neg p \wedge p^1)))$
 $= (\neg(p \wedge (\neg p \wedge \neg(p \wedge \neg p))) \wedge (p \wedge (\neg p \wedge \neg(p \wedge \neg p))))$

7.3 Inference and Consequence Relations

All systems $C0_1, \dots$ fall among modern systems, so their inference relations are just special cases of **definition 3.31**.

As for the contents of their consequence relations, there are two principles in question: the law of non contradiction

ion (LNC) and *ex contradictione sequitur quodlibet* (ECQ):

$$/ \neg(A \wedge \neg A) \quad (\text{LNC})$$

$$A, \neg A / B \quad (\text{ECQ})$$

While ECQ is restricted in all systems of paraconsistent logic, LNC will not be restricted in some of them.

It is worth noting that, in Graham Priest's LP system, LNC is valid in general. The underlying philosophy of LP is *dialtheism*, the thesis that there are true contradictions. That is, some expressions will be regarded as true, which are of the form:

$$(A \wedge \neg A) \quad (\neg)$$

Some of those will be precisely expressions of the form above. Thus, say that ' p ' stands for an expression such that ' $(p \wedge \neg p)$ ' is true. Thus, we see that $(\neg)$ holds for ' p '. Moreover, there would be no problem, under this framework, with saying that $(\neg)$ also holds for ' $(p \wedge \neg p)$ '. Hence, we would have that:

$$((p \wedge \neg p) \wedge \neg(p \wedge \neg p)). \quad (\forall)$$

From this, both ' $(p \wedge \neg p)$ ' and ' $\neg(p \wedge \neg p)$ ' would obtain. Thus, under this framework, LNC can safely be said to hold. ' p ' even when it is such that ' $(p \wedge \neg p)$ ' is the case.

For the rest, we want to have as many of the principles of standard logic as we can have, without having to include these principles.

A couple of problematic principles, in this respect, are the principle of addition ($\vee_1^+$ and $\vee_2^+$) and disjunctive syllogism (DS1 and DS2).

a	$(A \vee B)$		
	$\vdots$		
b	$\neg A$		
	$\vdots$		
	B	DS1, a, b	

a	$(A \vee B)$		
	$\vdots$		
b	$\neg B$		
	$\vdots$		
	A	DS2, a, b	

These principles are derivable in S0, and have a long-standing tradition in Western logic. The reason these principles are problematic in paraconsistent logics is that they easily allow to obtain ECQ as a derived rule.

Theorem 7.3. $\mathbf{DBas}_S = \{\vee_1^+, \vee_2^+, \text{DS1}, \text{DS2}\} \Rightarrow \exists \mathbf{D} : \mathbf{D} \text{ is a Fitch derivation on } \mathbf{Fo}_S \text{ from } \mathbf{DBas}_S \text{ with } A, \ulcorner \neg A \urcorner \in \mathbf{Prem}_S(\mathbf{D}) \text{ and } B \in \mathbf{Conc}_S(\mathbf{D}).$

Proof. Let $A, B \in \mathbf{Fo}_S$ and $\mathbf{DBas}_S = \{\vee_1^+, \vee_2^+, \text{DS1}, \text{DS2}\}$. Consider the schematic derivation:

1	A	
2	$\neg A$	
3	$(A \vee B)$	$\vee_1^+, 1$
4	B	DS2, 3, 2

Call this schematic derivation ‘**D**’. It is clear, for all $A, B \in \mathbf{Fo}_S$, that **D** is a Fitch derivation on $\mathbf{Fo}_S$ from $\mathbf{DBas}_S$. This means that there is a derivation $\mathbf{D} \in \mathbf{Der}_S$ with $A, \neg A \in \mathbf{Prem}_S(\mathbf{D})$ and $B \in \mathbf{Conc}_S(\mathbf{D})$. $\square$

We have only used $\vee_1^+$ and DS2, but we could have alternatively used $\vee_2^+$ and DS1. Hence, a paraconsistent system will normally have to restrict some of the principles $\vee_1^+, \vee_2^+, \text{DS1}, \text{DS2}$.

7.4 Interpretation System for $\mathbf{C0}_n$

The semantics for da Costa’s $\mathbf{C0}_1$ systems uses cpi (classical propositional interpretations), thus assigning to each propositional variable a classical truth value, either **1** (truth) or **0** (falsity).

A cpi $\mathfrak{I}$ is extended to all formulae of $\mathbf{Fo}_{\mathbf{C0}_1}$ by means of a valuation $\hat{\mathfrak{I}}$, which is its $\mathbf{C0}_1$ -extension, or $\mathbf{C0}_1\text{e}$, for short.

Definition 7.4 ($\mathbf{C0}_1$ -valuation). A valuation $\hat{\mathfrak{I}} = \langle \{\mathbf{0}, \mathbf{1}\}, \hat{\iota} \rangle$ is a **dCe** of the cpi $\mathfrak{I}$ on $\mathbf{Fo}_S$

$$\begin{aligned}
 &\stackrel{\text{DE}}{=} \hat{\iota}(\neg A) = \mathbf{0} \Rightarrow \hat{\iota}(A) = \mathbf{1} && \mathfrak{M} \\
 &\hat{\iota}(\neg \neg A) = \mathbf{1} \Rightarrow \hat{\iota}(A) = \mathbf{1} && \mathfrak{M} \\
 &\hat{\iota}(\neg B) = \hat{\iota}(\neg(A \rightarrow B)) = \hat{\iota}(\neg(A \rightarrow \neg B)) = \mathbf{1} \Rightarrow \hat{\iota}(\neg A) = \mathbf{0} && \mathfrak{M} \\
 &\hat{\iota}(A \wedge B) = \mathbf{1} \Leftrightarrow \hat{\iota}(A) = \mathbf{1} = \hat{\iota}(B) && \mathfrak{M} \\
 &\hat{\iota}(A \vee B) = \mathbf{1} \Leftrightarrow \hat{\iota}(A) = \mathbf{1} \vee \hat{\iota}(B) = \mathbf{1} && \mathfrak{M} \\
 &\hat{\iota}(A \rightarrow B) = \mathbf{1} \Leftrightarrow \hat{\iota}(A) = \mathbf{0} \vee \hat{\iota}(B) = \mathbf{1} && \mathfrak{M} \\
 &\hat{\iota}(A \heartsuit B) = \mathbf{0} \Rightarrow \hat{\iota}(A) = \mathbf{0} \vee \hat{\iota}(B) = \mathbf{0} \\
 &: \heartsuit \in \{\wedge, \vee, \rightarrow\} \mathfrak{M} A, B \in \mathbf{Fo}_S.
 \end{aligned}$$

7.5 Derivation Systems for $\mathbf{C0}_n$

As with intuitionistic logics, it is a good idea to start with axiomatic systems. In particular, an axiomatic derivation system for $\mathbf{C0}_1$ can be defined using the following inference forms as the rules of its deductive base:

$(A \rightarrow B), A \ / \ B$	(MPP)
$\ / \ (A \rightarrow (B \rightarrow A))$	(P1)
$\ / \ (((A \rightarrow B) \rightarrow ((A \rightarrow (B \rightarrow C)) \rightarrow (A \rightarrow C)))$	(P2)
$\ / \ (A \rightarrow (B \rightarrow (A \wedge B)))$	(P3)
$\ / \ ((A \wedge B) \rightarrow A)$	(P4)
$\ / \ ((A \wedge B) \rightarrow B)$	(P5)
$\ / \ (A \rightarrow (A \vee B))$	(P6)
$\ / \ (B \rightarrow (A \vee B))$	(P7)
$\ / \ (((A \rightarrow C) \rightarrow ((B \rightarrow C) \rightarrow ((A \vee B) \rightarrow C)))$	(P8)
$\ / \ (A \vee \neg A)$	(Tertium)
$\ / \ (\neg \neg A \rightarrow A)$	(DN2)
$\ / \ (B^1 \rightarrow ((A \rightarrow B) \rightarrow ((A \rightarrow \neg B) \rightarrow \neg A)))$	(C0 ₁ 1)
$\ / \ (A^1 \rightarrow (\neg A)^1)$	(C0 ₁ 2)
$\ / \ ((A^1 \wedge B^1) \rightarrow (A \wedge B)^1)$	(C0 ₁ 3)
$\ / \ ((A^1 \wedge B^1) \rightarrow (A \vee B)^1)$	(C0 ₁ 4)
$\ / \ ((A^1 \wedge B^1) \rightarrow (A \rightarrow B)^1)$	(C0 ₁ 5)

As before, we assume, in the context of these rules, that $\ulcorner(A \leftrightarrow B)\urcorner$ is an abbreviation of $\ulcorner((A \rightarrow B) \wedge (B \rightarrow A))\urcorner$.

Note that we have all axioms of positive logic, plus two axioms satisfied by S0 which are not satisfied by I0: (Tertium) and (DN2).

As for the rest of $C0_n$ systems, their axioms are very similar. We just change the last four with their corresponding $C0_n$ -versions, which are as follows:

$\ / \ (B^n \rightarrow ((A \rightarrow B) \rightarrow ((A \rightarrow \neg B) \rightarrow \neg A)))$	(C0 _n 1)
$\ / \ (A^n \rightarrow (\neg A)^n)$	(C0 _n 2)
$\ / \ ((A^n \wedge B^n) \rightarrow (A \wedge B)^n)$	(C0 _n 3)
$\ / \ ((A^n \wedge B^n) \rightarrow (A \vee B)^n)$	(C0 _n 4)
$\ / \ ((A^n \wedge B^n) \rightarrow (A \rightarrow B)^n)$	(C0 _n 5)

It is important to note that the larger the n , the weaker is the system. In other words, if we define the consequence relations of $C0_n$ based on the axiom schemata above, we will have:

$$\mathbf{Cn}_{C0_1} \supset \mathbf{Cn}_{C0_2} \supset \dots \supset \mathbf{Cn}_{C0_\omega}.$$

The system $C0_\omega$ – which we will not study here – results from applying the axioms above to expressions of the form $\ulcorner A^\omega \urcorner$, which results from applying the superindex ‘1’ infinitely many times. That is, we are not dealing with a system describable with the language S0, but with an infinitary language, i.e., a language containing formulae of infinite length.

Constructing natural derivation systems for C0_n systems is a bit more complicated than with the previous systems. To do so, we work, again, from the rules in $\mathbf{DBas}_{\text{P0}}$, and we add some which will allow us to obtain the additional results.

As the axioms show, both *tertium non datur* and double negation work, which means that we do not need many of the restrictions of intuitionistic logic. In particular, $\neg^+$ and $\neg^-$ will be rules in all C0_n systems. We also want to admit $\perp^-$, as we will want to apply the ECQ to some restricted set of formulae.

We nevertheless want to restrict $\perp^+$. In particular, ' $\perp$ ' will not be obtainable from any contradiction, but just from contradictions obtained of well-behaved formulae. Thus, we want the following rule to hold for a system named $\ulcorner \text{C0}_n \urcorner$.

Falsum-introduction.

$$\begin{array}{c|c} a & A^n \\ & \vdots \\ b & A \\ & \vdots \\ c & \neg A \\ & \vdots \\ & \perp \quad \perp^{n+}, a, b, c \end{array}$$

Furthermore, we introduce the following *propagation rules*, which allow us to infer that formulae composed from well-behaved formulae are themselves well-behaved.

Propagation rules.

$$\begin{array}{c|c} a & A^n \\ & \vdots \\ b & B^n \\ & \vdots \\ & (A \heartsuit B)^n \quad \heartsuit^{n+}, a, b \end{array} \qquad \begin{array}{c|c} a & A^n \\ & \vdots \\ & (\neg A)^n \quad \neg^{n+}, a \end{array}$$

$\heartsuit \in \{\wedge, \vee, \rightarrow, \leftrightarrow\}$

The rules $\wedge^{n+}, \vee^{n+}, \rightarrow^{n+}, \leftrightarrow^{n+}$ defined here are to be intuitively understood as follows: if A is well-behaved and B is well-behaved, then so are $\ulcorner (A \wedge B) \urcorner, \ulcorner (A \vee B) \urcorner, \ulcorner (A \rightarrow B) \urcorner, \ulcorner (A \leftrightarrow B) \urcorner$. Note, however, that $A, B, \ulcorner (A \wedge B) \urcorner, \ulcorner (A \vee B) \urcorner, \ulcorner (A \rightarrow B) \urcorner, \ulcorner (A \leftrightarrow B) \urcorner$ are not themselves derivable by this rule.

Finally, we need to add the proof by cases rules, which corresponds to the axiom (**Tertium**).

Proof by cases rule

a		A
		$\vdots$
b		B
		$\vdots$
c		$\neg A$
		$\vdots$
d		B
		$\vdots$
		$B \text{ PC, } a-b, c-d$

The system above suffices for characterising all systems $\mathbf{Cn}_{C0_n}$. Thus, we might define $\mathbf{DBas}_{C0_n}$ as follows:

Definition 7.5 (Deductive base for $C0_n$).

$$\mathbf{DBas}_{C0_n} \stackrel{\text{DF}}{=} \mathbf{DBas}_{P0} \cup \{\neg^-, \neg^+, \perp^-, \perp^{n+}, \neg^{n+}, \wedge^{n+}, \vee^{n+}, \rightarrow^{n+}, \leftrightarrow^{n+}, \text{PC}\}.$$

As for the derivation system $\mathbf{Cal}_{C0_n}$, we will assume that the functions $\mathbf{Prem}_{C0_n}$ and $\mathbf{Conc}_{C0_n}$ map each Fitch derivation with their premises and conclusions, respectively. Thus, we have the following definition.

Definition 7.6 ($\mathbf{Cal}_{C0_n}$).

$\mathbf{Cal}_{C0_n}$ is a calculus on $C0_n$

$$\begin{aligned} &\stackrel{\text{DF}}{=} \mathbf{Cal}_{C0_n} = \langle \mathbf{DBas}_{C0_n}, \mathbf{Der}_{C0_n}, \mathbf{Prem}_{C0_n}, \mathbf{Conc}_{C0_n}, \vdash_{C0_n} \rangle \\ &: \mathbf{DBas}_{C0_n} \text{ is a deductive base on } \mathbf{Fo}_{C0_n} \text{ } \mathbb{M} \\ &\mathbf{Der}_{C0_n} = \{D : D \text{ is a Fitch derivation on } \mathbf{Fo}_{C0_n} \text{ from } \mathbf{DBas}_{C0_n}\} \text{ } \mathbb{M} \\ &\mathbf{Prem}_{C0_n} : \mathbf{Der}_{C0_n} \rightarrow \wp \mathbf{Fo}_{C0_n} \text{ } \mathbb{M} \\ &\mathbf{Conc}_{C0_n} : \mathbf{Der}_{C0_n} \rightarrow \wp \mathbf{Fo}_{C0_n} \text{ } \mathbb{M} \\ &\vdash_{C0_n} = \{\langle A, B \rangle : \exists X \in \mathbf{Der}_{C0_n} (\mathbf{Prem}_{C0_n}(X) = A \text{ } \mathbb{M} \text{ } \mathbf{Conc}_{C0_n}(X) = B)\}. \end{aligned}$$

Task 7.2. Show that all inferences of the forms (**Tertium**), (**DN2**), (**C0_n1**)–(**C0_n4**) are members of $\vdash_{C0_n}$.

Theorem 7.7. $\vdash_{\mathbf{Co}_n} (\neg\neg A \rightarrow A)$.

Proof. Let $A \in \mathbf{Fo}_{\mathbf{Co}_n}$ and consider the schematic derivation:

1			$\neg\neg A$	
2			$(A \wedge \neg A)$	
3			A	$\wedge_1^-, 2$
4			$\neg(A \wedge \neg A)$	
			A^n	Abbr., 4
5			$(\neg A)^n$	$\neg^{n+}, 4$
6				$\neg A$
7				$\perp$
				$\perp^{n+}, 5, 6, 1$
8			A	$\neg^-, 6-7$
9			A	PC, 2-3, 4-8
10			$(\neg\neg A \rightarrow A)$	$\rightarrow^+, 1-9$

Call this schematic derivation ‘**D**’. It is clear, for all $A \in \mathbf{Fo}_{\mathbf{Co}_n}$, that **D** is a Fitch derivation on $\mathbf{Fo}_{\mathbf{Co}_n}$ from $\mathbf{DBas}_{\mathbf{Co}_n}$. This means that there is a derivation $\mathbf{D} \in \mathbf{Der}_{\mathbf{Co}_n}$ with $\{\} \subseteq \mathbf{Prem}_{\mathbf{Co}_n}(\mathbf{D})$ and $\ulcorner (\neg\neg A \rightarrow A) \urcorner \in \mathbf{Conc}_{\mathbf{Co}_n}(\mathbf{D})$. Therefore, $\vdash_{\mathbf{Co}_n} (\neg\neg A \rightarrow A)$ has to hold according to [definition 7.6](#). $\square$

Theorem 7.8. $\vdash_{\mathbf{C0}_n} (B^n \rightarrow ((A \rightarrow B) \rightarrow ((A \rightarrow \neg B) \rightarrow \neg A)))$.

Proof. Let $A, B \in \mathbf{Fo}_{\mathbf{C0}_n}$ and consider the schematic derivation:

1			B^n				
2				$(A \rightarrow B)$			
3					$(A \rightarrow \neg B)$		
4						A	
5						B	$\rightarrow^-, 2, 4$
6						$\neg B$	$\rightarrow^-, 3, 4$
7						$\perp$	$\perp^{n+}, 1, 5, 6$
8						$\neg A$	$\neg^+, 4-7$
9					$((A \rightarrow \neg B) \rightarrow \neg A)$	$\rightarrow^+, 3-8$	
10				$((A \rightarrow B) \rightarrow ((A \rightarrow \neg B) \rightarrow \neg A))$	$\rightarrow^+, 2-9$		
11		$(B^n \rightarrow ((A \rightarrow B) \rightarrow ((A \rightarrow \neg B) \rightarrow \neg A)))$	$\rightarrow^+, 1-11$				

Call this schematic derivation 'D'. It is clear, for all $A, B \in \mathbf{Fo}_{\mathbf{C0}_n}$, that D is a Fitch derivation on $\mathbf{Fo}_{\mathbf{C0}_n}$ from $\mathbf{DBas}_{\mathbf{C0}_n}$. This means that there is a derivation $\mathbf{D} \in \mathbf{Der}_{\mathbf{C0}_n}$ with $\{\} \subseteq \mathbf{Prem}_{\mathbf{C0}_n}(\mathbf{D})$ and $\ulcorner (B^n \rightarrow ((A \rightarrow B) \rightarrow ((A \rightarrow \neg B) \rightarrow \neg A))) \urcorner \in \mathbf{Conc}_{\mathbf{C0}_n}(\mathbf{D})$. Therefore, $\vdash_{\mathbf{C0}_n} ((A \rightarrow B) \rightarrow ((A \rightarrow (B \rightarrow C)) \rightarrow (A \rightarrow C)))$ has to hold according to [definition 7.6](#). $\square$

Note also that, in $\mathbf{C}_n$, the strong negation ' $\neg^{(n)}$ ' has the properties of classical negation.

Task 7.3. Show that all inferences of the following forms are members of $\vdash_{\mathbf{C0}_n}$.

1. $/ (A \vee \neg^{(n)} A)$
2. $/ \neg^{(n)}(A \wedge \neg^{(n)} A)$
3. $(A \rightarrow B), (A \rightarrow \neg^{(n)} B) / \neg^{(n)} A$
4. $A, \neg^{(n)} A / B$
5. $A / \neg^{(n)} \neg^{(n)} A$
6. $(A \leftrightarrow \neg^{(n)} A) // B$

Theorem 7.9. $A, \overset{(n)}{\neg} A \vdash_{C0_n} B$.

Proof. Let $A \in \mathbf{Fo}_{C0_n}$ and consider the schematic derivation:

1	A	
2	$\overset{(n)}{\neg} A$	
	$(\neg A \wedge A^{(n)})$	Abbr., 2
3	$\neg A$	$\wedge_1^-, 2$
4	$A^{(n)}$	$\wedge_2^-, 2$
5	$\perp$	$\perp^{n+}, 4, 1, 3$
6	B	$\perp^-, 5$

Call this schematic derivation '**D**'. It is clear, for all $A, B, C \in \mathbf{Fo}_{C0_n}$, that **D** is a Fitch derivation on $\mathbf{Fo}_{C0_n}$ from $\mathbf{DBas}_{C0_n}$. This means that there is a derivation $\mathbf{D} \in \mathbf{Der}_{C0_n}$ with $A, \overset{(n)}{\neg} A \in \mathbf{Prem}_{C0_n}(\mathbf{D})$ and $B \in \mathbf{Conc}_{C0_n}(\mathbf{D})$. Therefore, $A, \overset{(n)}{\neg} A \vdash_{C0_n} B$ has to hold according to [definition 7.6](#). $\square$